

# AI Fact-Checking System

Group ID  
COMP90042 Project 2026

## Abstract

We present a resource-aware climate fact-checking system optimized for limited compute, with a standard Colab-style runtime as the concrete constraint. The pipeline stages evidence processing: sparse fusion preserves broad recall, embedding plus factual-cue reranking compresses the pool, a bounded cross-encoder pass selects submitted evidence, and a TF-IDF logistic classifier predicts the verdict from structured evidence text. This Colab-feasible design obtains 0.2021 evidence F-score, 0.5000 claim accuracy, 0.2879 harmonic mean, and 0.4659 classifier macro-F1.

## 1 Introduction

Climate claims are short, compressed, and evidence-sensitive; a useful fact-checker must retrieve evidence as well as predict a label. Practical systems also face limited compute, so full-corpus neural scoring is often unavailable. We instantiate this constraint with a Colab-style runtime: no external services, saved checkpoints, or precomputed ranked artifacts. The problem is therefore a three-way trade-off among evidence recall, verdict accuracy, and runtime.

**Draft note:** I collected and grouped the following papers for teammates to continue the related-work writing.

Existing work can be organized into four areas:

1. **Evidence-Based Fact Checking** FEVER frames fact checking as joint evidence retrieval and verdict prediction (Thorne et al., 2018); CLIMATE-FEVER adds climate-specific domain and retrieval challenges (Diggelmann et al., 2020).
2. **Large-Scale Evidence Retrieval** BM25 is fast and scalable (Robertson et al., 1994); reciprocal rank fusion combines ranked lists without supervision (Cormack et al., 2009).

These methods suit high-recall candidate generation but remain lexical.

3. **Neural Semantic Reranking** Dense embeddings help when lexical overlap is weak (Karpukhin et al., 2020; Reimers and Gurevych, 2019); cross-encoders model claim–evidence interaction directly (Nogueira and Cho, 2019), but their cost requires sparse prefiltering.
4. **Evidence Aggregation and Factual Cue Fusion** Evidence aggregation studies passage combination before classification (Zhou et al., 2019). Hybrid systems show lexical, semantic, and shallow factual signals can be complementary (Lee et al., 2018; Padia et al., 2018).

Our design is literature-aligned but resource-driven. It first builds a broad unsupervised sparse pool from word, character, structured, and PRF gates. It then compresses this pool with MiniLM similarity plus factual features, applies a bounded cross-encoder only after prefiltering, and classifies from rank-aware, cue-aware evidence text. Contributions: (i) a staged pipeline for the performance–runtime trade-off, (ii) multi-gate sparse retrieval, (iii) multi-scale reranking that fuses semantic and shallow factual signals, and (iv) diagnostics over sparse gates, fusion, evidence selection, and classifier sensitivity.

## 2 Method

### 2.1 Pipeline Overview

Let  $\mathcal{C}$  be claims and  $\mathcal{E}$  evidence. The output is a verdict and a small evidence set per claim. Since scoring all  $(c, e) \in \mathcal{C} \times \mathcal{E}$  pairs is infeasible, we use a staged pipeline:

$$\mathcal{E} \rightarrow \mathcal{A}(c) \rightarrow \mathcal{B}(c) \rightarrow \mathcal{D}(c),$$

where  $\mathcal{A}(c)$  is a broad sparse pool,  $\mathcal{B}(c)$  a reranked context pool, and  $\mathcal{D}(c)$  the submitted evidence.

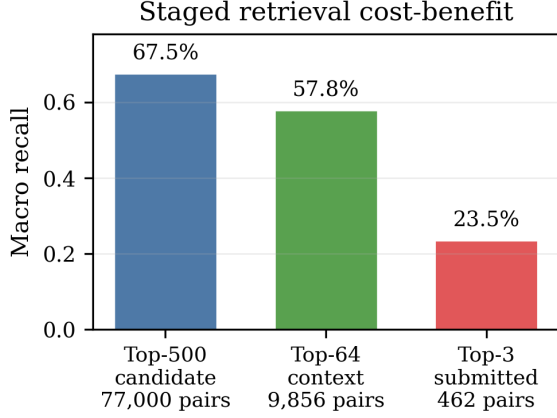


Figure 1: Staged pruning keeps recall while reducing pair count. The candidate and context pools retain most recoverable evidence without full-corpus neural scoring, which would require about 186 million development pairs.

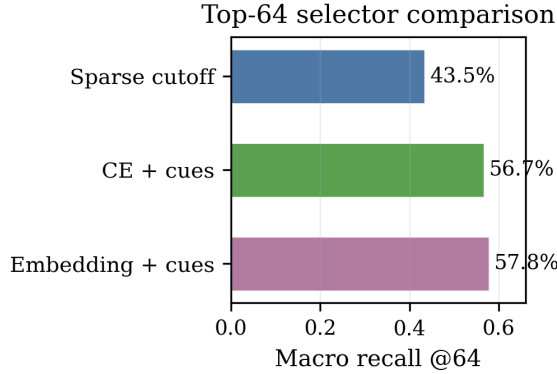


Figure 2: Embedding plus factual cues is the best top-64 selector, reaching 57.8% macro recall above sparse cutoff and cross-encoder-plus-cue variants.

Cheap stages maximize recall; expensive semantic models operate only after pruning. The four components are multi-gate sparse retrieval, multi-scale reranking, bounded cross-encoder selection, and structured-evidence classification.

## 2.2 Multi-Gate Sparse Candidate Retrieval

The sparse stage maximizes recall. A gate is a feature construction, not a new scoring formula: the four gates differ in analyzer  $a_g(\cdot)$  and vocabulary  $V_g$ , while sharing BM25. Let  $q_g(c)$  and  $x_g(d)$  be the resulting count vectors.

**Word gate.** Word terms give the direct lexical view: content-word overlap raises scores and IDF discounts frequent, weakly discriminative words.

**Character gate.** Character n-grams capture partial overlap, inflection, hyphenation, and spelling variation missed by exact words.

**Structured gate.** Structured cues keep numbers, years, salient capitalized terms, and negation markers. Standalone recall is low, but the cues add factual constraints that lexical views may lose.

**Pseudo-relevance feedback gate.** PRF expands the word query using high-ranked feedback evidence. If  $\mathcal{F}(c)$  is the feedback set,

$$q_{\text{prf}}(c) = q_{\text{word}}(c) + \eta \sum_{d \in \mathcal{F}(c)} w(c, d) x_{\text{word}}(d),$$

where  $w(c, d)$  downweights lower-ranked feedback documents. PRF adds related terms absent from the original claim while staying in the sparse word space.

**Shared BM25 interaction.** After feature construction, each gate applies the same BM25 interaction. For term  $t$  in document  $d$ ,

$$\text{idf}_t = \log \left( \frac{N - df_t + 0.5}{df_t + 0.5} + 1 \right),$$

$$\text{BM25}_{d,t} = \text{idf}_t \frac{(k_1 + 1) f_{d,t}}{f_{d,t} + k_1 (1 - b + b |d| / |d|)}.$$

IDF is part of BM25, not a separate gate; it discounts high-frequency features. The query side remains a count vector, giving the sparse dot product

$$s_g(c, d) = q_g(c)^\top \text{BM25}_g(d).$$

Thus  $g$  chooses the representation, and BM25 weights it. The notebook uses SciPy/NumPy sparse matrices, so document-frequency counts, BM25 reweighting, and query–document dot products run in optimized C/C++ kernels and vectorized array operations.

**Rank fusion.** Each gate produces a ranked list. The fifth sparse view is their weighted RRF fusion:

$$S_{\text{RRF}}(c, d) = \sum_{g \in G} \frac{\lambda_g}{K_{\text{RRF}} + r_g(c, d)},$$

where missing documents contribute zero. We treat  $\lambda_g$  as a reliability prior: stronger standalone gates receive larger weights, while weak but complementary signals can retain small non-zero weights. Structured recall is low alone, yet ablation shows it improves fused top-500 recall; RRF also avoids cross-gate score calibration.

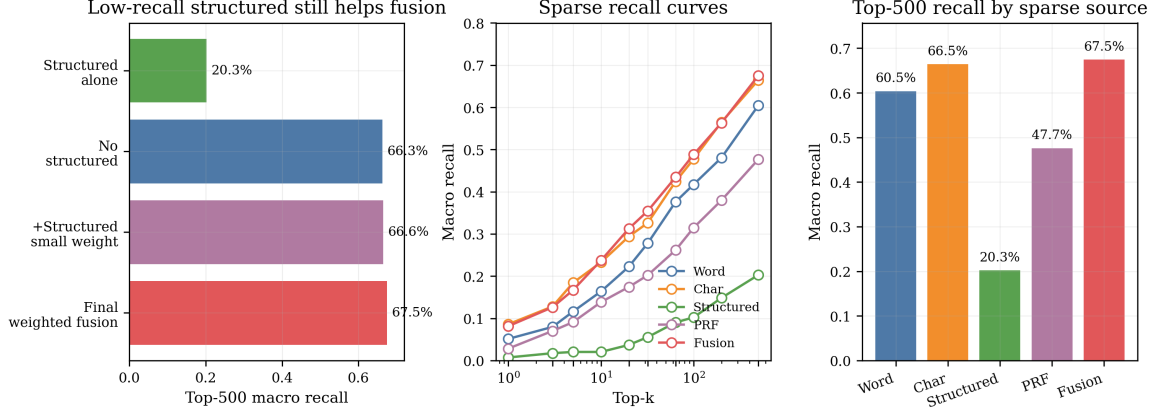


Figure 3: Sparse fusion gives the best top-500 recall and retains structured cues for complementarity. Character matching is the strongest single source; structured cues are weak alone, but removing them lowers fused recall under a controlled ablation, so the final system keeps a small structured weight.

Figures 1 and 3 justify the candidate stage: fusion improves top-500 recall over any single source, and staging avoids full-corpus neural scoring.

### 2.3 Multi-Scale Feature Fusion

Reranking fuses dense similarity, sparse source rank, word overlap, numeric agreement, negation compatibility, and length shape. MiniLM embeddings are mean-pooled and  $\ell_2$ -normalized, so cosine similarity is an inner product:

$$s_{\text{emb}}(c, d) = h(c)^\top h(d).$$

The shallow factual features are summarized as

$$\phi(c, d) = [\phi_{\text{rank}}, \phi_{\text{lex}}, \phi_{\text{num}}, \phi_{\text{neg}}, \phi_{\text{len}}, s_{\text{emb}}].$$

where the components encode source rank, lexical overlap, number match, negation compatibility, length mismatch, and embedding similarity. A lightweight gradient-boosted matcher estimates  $p_{\text{feat}}(c, d)$ , and the context score is

$$S_{\text{ctx}}(c, d) = \alpha \tilde{s}_{\text{emb}}(c, d) + (1 - \alpha) p_{\text{feat}}(c, d),$$

where  $\tilde{s}_{\text{emb}}$  is claim-wise normalized. Embeddings handle semantic relatedness; factual features guard against number, negation, and weak lexical-grounding errors.

### 2.4 Bounded Neural Evidence Selection

The submitted evidence stage uses a stricter operating point than the classifier context. Because cross-encoders jointly encode claim and evidence but are expensive, we apply them only after sparse/embedding prefiltering and normalize their scores within each claim. The final evidence score

fuses cross-encoder confidence, embedding confidence, embedding rank, and sparse source rank:

$$S_{\text{sub}}(c, d) = \beta_1 \tilde{s}_{\text{CE}} + \beta_2 \tilde{s}_{\text{emb}} + \frac{\beta_3}{\tau + r_{\text{emb}}} + \frac{\beta_4}{\tau + r_{\text{src}}}.$$

Coefficients are fixed. Candidates outside the cross-encoder subset receive no CE contribution but can still be selected by embedding and source-rank terms, bounding neural cost without discarding useful sparse candidates.

Figures 2 and 4 justify multi-scale fusion: factual cues are not standalone rules, but complements to semantic similarity.

### 2.5 Structured Evidence Classification

The classifier predicts from a bounded ranked evidence context. Rather than an unordered passage block, it constructs

$$x_c = [\text{claim}; z_1, z_1, \dots, z_m, z_m, z_{m+1}, \dots, z_L],$$

where top evidence is repeated to raise TF-IDF weight. Each evidence line also gets a rank prefix plus word-overlap, numeric-match, and negation-match cue tokens. These are learnable tokens, not rules. Let  $v_c$  be the TF-IDF vector; a one-vs-rest logistic model learns one binary classifier per label:

$$P(y = k | c) = \sigma(\theta_k^\top v_c + b_k).$$

The largest decision score gives the label. This keeps classification cheap while preserving retrieval rank and factual signals.

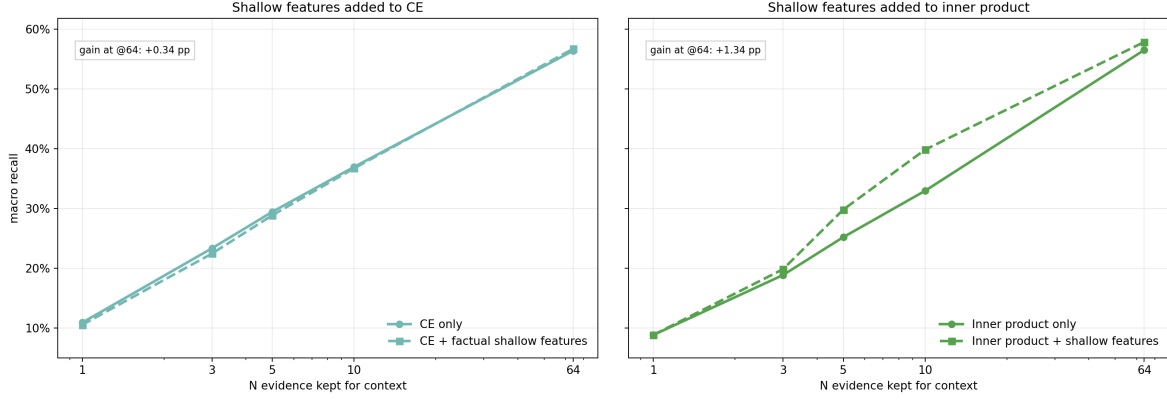


Figure 4: Factual cues preserve or improve semantic ranking. Gains are small for the strong cross-encoder but clearer for embedding inner product, supporting feature fusion over a purely semantic selector.

| Stage           | F-score | Macro recall | Hit-any |
|-----------------|---------|--------------|---------|
| Sparse top-500  | 0.0083  | 0.6754       | 0.9026  |
| Context top-64  | 0.0517  | 0.5784       | 0.8636  |
| Submitted top-3 | 0.2021  | 0.2355       | 0.4675  |

Table 1: Retrieval metrics from the self-generated notebook artifacts.

| Metric                                | Value  |
|---------------------------------------|--------|
| Evidence Retrieval F-score ( $F$ )    | 0.2021 |
| Claim Classification Accuracy ( $A$ ) | 0.5000 |
| Harmonic Mean of $F$ and $A$ ( $H$ )  | 0.2879 |
| Top-3 Macro Recall                    | 0.2355 |
| Classifier Macro-F1                   | 0.4659 |

Table 2: Final Colab-reproduced development result.  $F$ ,  $A$ , and  $H = 2FA/(F + A)$  are official course metrics; the last two rows summarize retrieval and classification behavior.

### 3 Results

The final notebook was run in Colab with GPU enabled. Course metrics are evidence F-score  $F$ , label accuracy  $A$ , and H-score  $H = 2FA/(F + A)$ . Tables 1–2 report retrieval operating points and reproduced development results.

Figure 5 shows that remaining errors mainly confuse SUPPORTS, REFUTES, and NEI; DISPUTED has only 18 development examples.

Table 3 justifies the classifier input: enhanced context improves over plain context at  $C = 0.25$  (0.4935/0.4544 vs. 0.4351/0.3762), and the diagnostic best is  $C = 0.125$  (0.5195/0.4841). The reported Colab run is Table 2:  $F = 0.2021$ ,  $A = 0.5000$ ,  $H = 0.2879$ .

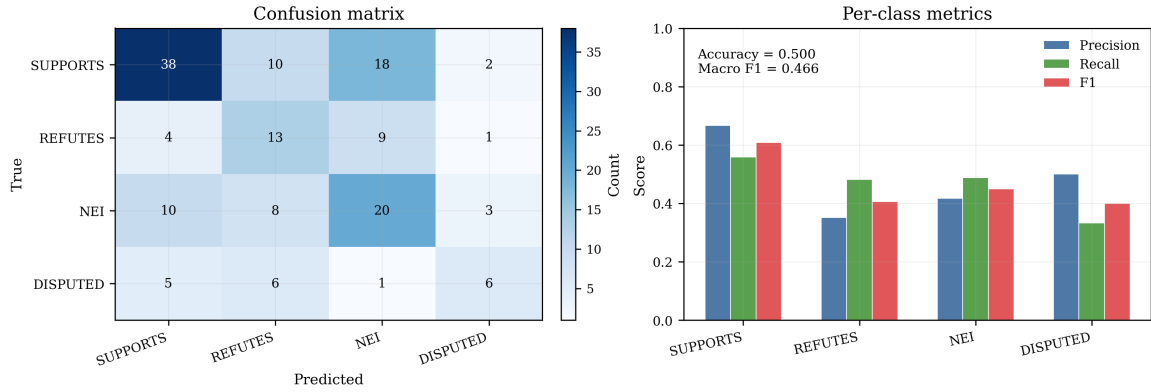


Figure 5: Classifier errors concentrate among the three main labels. The final Colab run is strongest on SUPPORTS; DISPUTED remains hard because it is rare and often evidence-conflict dependent.

| Context format | $C$   | Acc.   | Macro-F1 |
|----------------|-------|--------|----------|
| Plain          | 0.25  | 0.4351 | 0.3762   |
| Rank-aware     | 0.25  | 0.4351 | 0.3749   |
| Top-weighted   | 0.25  | 0.4416 | 0.3938   |
| Rank-weighted  | 0.25  | 0.4351 | 0.3882   |
| Enhanced       | 0.25  | 0.4935 | 0.4544   |
| Enhanced       | 0.125 | 0.5195 | 0.4841   |

| $C$   | Acc.   | Macro-F1 |
|-------|--------|----------|
| 0.125 | 0.5195 | 0.4841   |
| 0.25  | 0.4935 | 0.4544   |
| 0.5   | 0.4740 | 0.4263   |
| 1.0   | 0.4675 | 0.4184   |
| 2.0   | 0.4416 | 0.3825   |
| 4.0   | 0.4545 | 0.3913   |
| 8.0   | 0.4481 | 0.3925   |

Table 3: Classifier ablations. Rank-aware, top-weighted, factual-cue-enhanced context improves over plain concatenation; the enhanced classifier peaks at  $C = 0.125$  in the diagnostic sweep.

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